Circuits Review — Capacitors & Kirchhoff's Rules

The loop rule applies to a capacitor just like it does to a resistor or a battery — you just need $\Delta V_C = Q/C$ for whatever term the capacitor contributes. Part A uses that at some instant *during* charging; Part B uses it once a circuit has reached steady state, when a fully charged capacitor carries no current.

Part A — Charge at an Instant (RC Charging)

1. A capacitor C is charging through a resistor R, connected to a battery ε, all in a single loop. At a certain moment, while charge is still building up, the current through the resistor is I.

$$\varepsilon = 9.0\ \text{V} \qquad R = 2.0\ \text{k}\Omega \qquad C = 4.0\ \mu\text{F} \qquad I = 1.5\ \text{mA (at this moment)}$$

What is the charge Q on the capacitor at that same moment?

2. Two capacitors, $C_1 = 4.0\ \mu\text{F}$ and $C_2 = 6.0\ \mu\text{F}$, are wired in parallel with each other. That parallel pair is in series with a resistor $R = 2.0\ \text{k}\Omega$ and a battery $\varepsilon = 15\ \text{V}$ — so the full loop is battery $\to R \to$ the $C_1 \parallel C_2$ pair $\to$ back to the battery.

At a certain moment during charging, the charge on C_1 is measured to be $Q_1 = 20\ \mu\text{C}$.

Find (a) the voltage across the parallel pair at that moment, (b) the current through R at that moment, and (c) the charge Q_2 on C_2 at that same moment.

Part B — Steady-State Capacitor Voltage

3. A battery $\varepsilon = 10$ V is in series with two resistors, $R_1 = 1.5$ kΩ and $R_2 = 2.5$ kΩ, and a capacitor $C = 3.0$ μF — all in one single loop. The circuit has been connected for a long time (steady state). What is the charge Q on the capacitor?

4. A battery $\varepsilon = 12$ V is in series with a resistor $R_1 = 2.0$ kΩ. That combination feeds a parallel pair: a second resistor $R_2 = 4.0$ kΩ and a capacitor $C = 5.0$ μF (so R_2 and C are in parallel with each other, and that whole pair is in series with R_1 and the battery). The circuit has reached steady state.
Find (a) the current through R_1 (and through R_2), and (b) the charge Q on the capacitor.

ANSWER KEY — FLIP THE PAGE OVER (OR STAND ON YOUR HEAD)

Part A — Charge at an Instant (RC Charging)
1. $Q = 24\ \mu\text{C}$
2. (a) $\Delta V = 5.0\ \text{V}$ (b) $I = 5.0\ \text{mA}$ (c) $Q_2 = 30\ \mu\text{C}$

Part B — Steady-State Capacitor Voltage
3. $Q = 30\ \mu\text{C}$ — note R_1 and R_2 never mattered: at steady state $I = 0$ everywhere in this loop, so neither resistor drops any voltage, and $Q = C\varepsilon$ regardless of their values.
4. (a) $I = 2.0\ \text{mA}$ (b) $Q = 40\ \mu\text{C}$ — here the resistor values *do* matter, since R_2 gives current a path that doesn't go through C.